

Alexander Tong

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EDUCATION

Boston University

B.S. Mechanical Engineering, Aerospace Concentration

Expected May 2028

GPA: 3.71/4.00

PROFESSIONAL EXPERIENCE

Boston University Plummer Laboratory

Mechanical Engineering Assistant

June 2026 - Present

Boston, MA

- Designed a vibration test fixture in SolidWorks; predicted 0.44 mm deflection at 3 kg and 20.1 Hz using dynamic FEA and analyzed bolt preload, thread shear, plate bearing, and pull-through
- CNC-machined Al 6061-T6 shaker adapter via Haas VF-2 CNC Mill to a 0.005 in tolerance; generated G-code and engineering drawing
- Reduced fixture system cost from a \$1,170 outsourced estimate to \$71.11 (93.9% reduction) through in-house fabrication
- Designed the test matrix, procedure, and three of four specimen concepts for 48 Instron 68SC-5 tests; processed 0–181 mN force data to investigate physical Ising models

Culturion

Mechanical Engineering Intern

February 2026 - June 2026

Sydney, Australia

- Planned and executed the company's first full-scale nanoparticle contamination deposition mapping across chamber locations and magnetic configurations using SEM, optical microscopy, visual evidence, and ImageJ
- Presented pump-port/chamber and electron-curtain contamination-prevention concepts based on mapping and fluid/electromagnetic first principles in a technical design review
- Designed Onshape sheet-metal sample racks for reactor-port insertion and adhesive-free mounting

PROJECTS

Aerodynamic Downcomer Fairing

Project Lead, Boston University Rocket Propulsion Group

August 2026 - Present

- Leading set-based design of a fairing based in missile design philosophy to protect Argo vehicle's downcomer from thermal and aerodynamic loads; comparing three concepts for supersonic drag, loading, thermal effects, and geometric constraints
- Performing Ansys CFD at Mach 2.0–2.81 across candidates and applying adjoint optimization

Argo Separation and Recovery Subsystem

Co-Lead Engineer, Boston University Rocket Propulsion Group

June 2025 - Present

- Designed redundant recovery for a 135 lb dry-mass rocket targeting 100,000 ft through four design reviews; final subsystem mass 18.7 lbm within a 20 lbm budget, final BOM \$1,076
- Calculated 823 lbf parachute shock and 1,230 lbf snatch loads using five published methods
- Wrote standard operating procedures (SOPs) for system tests; led crew briefings, fire/abort calls, and post-test analysis
- Developed CO2 deployment hardware after integrated-test failures, performing pressure and thick-wall stress calculations, CAD, drawings, manual lathe machining, and component testing
- Built 352 L vacuum chamber operating at 0.011 atm with LabJack pressure acquisition testing CO2/BP deployment
- Dynamically tested Brummel eye splices up to 1.5 FOS; Projected 90–110% strength gain over knots from published data
- Redesigned and machined cable cutters, reducing mass 49.5% with a calculated 2.01 FOS

AIAA Design-Build-Fly Aircraft

Wing Structural Lead, Boston University

June 2025 - December 2025

- Led material, structural, and manufacturing trade studies for two Fusion 360 wing designs; produced an XPS prototype 50% lighter than the prior year's wing and cut build time from three months to one
- Mentored nine members and next wing design lead in Fusion 360, DFMA principles, and laser cutting

Hybrid Rocket Nozzle

Lead Engineer, Boston University Rocket Propulsion Group

September 2024 - November 2024

- Led five members through three design reviews and a two-second hotfire; reduced nozzle mass 73% from the initial design using SolidWorks CAD/FEA with a calculated 7.8 FOS under mechanical loads alone

TECHNICAL SKILLS

SolidWorks, Ansys CFD, FEA, CNC & Manual Machining, DFMA, Composite Layup, LabJack DAQ, Python